

Roll No.

TEC–200

B. Tech. (CSE) (Third Semester)

EXAMINATION, 2025-26

DIGITAL ELECTRONICS

Time : $2\frac{1}{2}$ Hours

Maximum Marks : 60

Note : Question paper has three Sections. Read carefully instructions for each Section.

Section—A

(Very Short Answer Type Questions)

1. Attempt all questions. 1 each

(a) What is don't care condition in K-map ?

(CO1, RE)

(b) Convert $(A5B.11)_{16} \rightarrow (?)_8$. (CO1, AP)

P. T. O.

- (c) Explain the De-Morgan's theorem. (CO1, RE)
- (d) What is fan-out in logic families ? (CO4, UN)
- (e) Describe Sequential circuit. (CO3, UN)
- (f) Differentiate between synchronous and asynchronous counter. (CO4, UN)
- (g) How does a full adder differ from a half adder ?
(CO2, UN)
- (h) Describe Read Only Memory. (CO5, UN)
- (i) Express the Truth Table of JK Flip Flop.
(CO3, UN)
- (j) Define a combinational circuit. (CO2, UN)
- (k) Name *one* application of multiplexers.
(CO2, RE)
- (l) What is the binary addition of $1011 + 0101$?
(CO1, AP)

Section—B

(Short Answer Type Questions)

- 2. Attempt all questions. 4 each
 - (a) Explain the working of TTL logic family with suitable truth tables and circuit diagram.
(CO4, UN)

OR

Explain various types of logic families with examples. (CO4, UN)

- (b) Solve following function using K-Map :

(CO1, AP)

$$F(A, B, C, D) = \Sigma m(1, 3, 7, 11, 15) + d(0, 2, 5)$$

OR

Using Quine-McCluskey method simplify the following expression : (CO1, AP)

$$F(A, B, C, D) = \Sigma m(0, 4, 5, 7, 8, 11, 12, 15)$$

- (c) Discuss and implement BCD adder with the help of logic gates. (CO2, CR)

OR

Design 3 to 8 Decoder circuit using logic gates. (CO2, CR)

- (d) Design JK flip flop using T Flip Flop.

(CO3, CR)

OR

Design S R flip flop using D Flip Flop.

(CO3, CR)

P. T. O.

- (e) Draw the circuit diagram of CMOS inverter. Draw its transfer characteristics and explain its operation. (CO4, CR)

OR

Design PMOS NOR Gate using MOSFET and explain its Working. (CO4, CR)

- (f) Describe the concept of programmable logic devices (PLDs) and explain how they are used for logic implementation in digital systems.

(CO5, UN)

OR

Explain the Programmable Array Logic with suitable example. (CO5, UN)

Section—C

(Long Answer Type Questions)

3. Attempt all questions. 8 each
- (a) Explain the working of JK Flip flop with all the required tables. (CO3, AP)

OR

Explain the working of SR Flip flop with all the required tables. (CO3, AP)

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- (b) Explain the working of 8 : 1 Multiplexer. Design the following Boolean Function using 4 : 1 Multiplexer $(A, B, C, D) = \Sigma m(1, 3, 5, 8, 11, 12, 14, 15)$. (CO2, EV)

OR

Describe the operation of half Subtractor and full Subtractor. Draw its truth tables and circuit diagrams. (CO2, EV)

- (c) Design 3 Bit down Counter using Flip Flops. (CO3, CR)

OR

Design a Shift right register using Flip Flops and explain its operation with timing diagram. (CO3, CR)

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